NMD 2026-04-14: Knee X-Ray Findings and Exercise Plan

Conversation with Gemini

User: Summarize the findings of the latest knee X-ray.

Then provide a targeted exercise strategy for gym training based on those findings:

1. Key biomechanical issues to address
2. Recommended exercise categories:

- Strength (specific muscles)
- Mobility
- Stability / neuromuscular control

3. Provide a list of specific exercises, each with:

- Purpose (what it addresses)
- Key technique cues
- Contraindications (if any)

4. Include YouTube video links demonstrating proper form

Assistant: *Nick's medical dialogs (NMD) — Custom Gem*

This is for informational purposes only. For medical advice or diagnosis, consult a professional.

File Inventory

File	Loaded	Parsed
imaging_and_procedures.md	Yes	Yes
medical_summary.md	Yes	Yes
medications_and_treatments.md	Yes	Yes
labs.csv	Yes	Yes

CSV File Details

File: labs.csv

Columns: Date, Category, Marker, Result, Units, Reference Range, Notes, Aliases

Row Count: 64

As an AI, I am acting as your clinical archivist to synthesize your records and build an evidence-based framework for your gym routine. Here is the summary of your right knee imaging and a targeted training strategy to keep you active and protect your joints.

1. Knee X-Ray Summary

Your latest right knee X-ray (3 views, weight-bearing) from 2024-04-22 indicates Grade 2 osteoarthritis, which is primarily localized to the medial (inner) tibiofemoral compartment.

Key Findings:

Medial Compartment: Shows moderate joint space narrowing (reduced to 3.2mm), marginal osteophyte (bone spur) formation, and mild subchondral sclerosis.

Lateral Compartment: Well-preserved with no significant degeneration.

Patellofemoral Compartment: Shows mild lateral facet osteophytes.

General: No acute fractures or dislocations.

The radiologist noted that these findings are consistent with age-related degenerative changes, likely accelerated by your consistently high activity level.

2. Targeted Exercise Strategy: Biomechanical Issues to Address

Based on your X-ray and your physical therapy discharge summary from June 2024, your gym programming needs to address three primary biomechanical factors:

Medial Compartment Overload: Because your degeneration is focused on the medial side, we need to prevent the knee from caving inward (knee valgus) during movement, which was a specific deviation noted by your physical therapist.

VMO / Quadriceps Weakness: Your PT identified significant quadriceps weakness, particularly in the vastus medialis oblique (VMO). The VMO is crucial for stabilizing the kneecap and absorbing shock.

Axial Loading Vulnerability: Your orthopedic recommendation explicitly advises avoiding high-impact axial loading like heavy leg presses and running. This also aligns with protecting your L5-S1 lumbar degenerative disc disease.

3. Recommended Exercise Categories

To keep you in the gym 5x a week safely, focus on:

Strength: Quadriceps (specifically VMO isolation) and Glutes (maximus and medius to prevent valgus collapse).

Mobility: Ankle dorsiflexion and hip mobility. If the joints above and below the knee are mobile, the knee can act as a stable hinge rather than absorbing unwanted rotational forces.

Stability / Neuromuscular Control: Single-leg movement patterns that challenge pelvic control and force proper knee tracking.

4. Specific Exercise Prescriptions

Here are joint-sparing exercises designed to build strength around the knee without subjecting it—or your lumbar spine—to heavy, compressive loads.

Terminal Knee Extensions (TKEs)

Purpose: Isolates and activates the quadriceps (specifically the VMO) to build strength and terminal extension control without requiring deep knee flexion.

Key Technique Cues: Anchor a heavy resistance band around a rig at knee height. Step inside the band so it rests just behind your right knee. Step back to create tension. Start with a bent knee, then powerfully squeeze your quad to push your knee straight back against the band. Hold the contraction for 2 seconds.

Contraindications: Discontinue if you feel sharp, pinching pain at the front of the kneecap.

Video Demonstration: How to perform the banded Terminal Knee Extension

Spanish Squats

Purpose: Highly effective for building quadriceps mass and tendon resilience while keeping your shins vertical. This minimizes patellofemoral shear forces and completely bypasses the heavy axial loading of a traditional barbell squat.

Key Technique Cues: Loop a heavy resistance band around a rig and step both legs in, placing the band behind both knees. Walk back until the band is fully tensioned. Sit back into a squat while leaning slightly back against the band. Keep your torso upright and shins perfectly vertical. Drive your knees back into the band as you stand up.

Contraindications: Avoid dropping below a 90-degree knee angle if it aggravates your patellofemoral osteophytes.

Video Demonstration: Spanish Squat Technique

Banded Lateral Walks

Purpose: Targets the gluteus medius to improve lateral hip stability. Strong hip abductors are the primary defense against the knee caving inward (valgus) under load, which protects your medial joint space.

Key Technique Cues: Place a resistance band just above your knees. Hinge slightly at the hips into an athletic stance. Step laterally, maintaining tension on the band at all times (do not let the feet come completely together). Keep your toes pointed straight forward—do not let the lead foot externally rotate.

Contraindications: Ensure the movement comes from the hips, not by tilting or swaying the lower back, to protect your L5-S1 disc.

Video Demonstration: Banded Lateral Walk
